Research Report: EcoSort AI — Technical Specification and Product Strategy

1. Executive Summary

The rapid urbanization and economic expansion of India have precipitated a municipal solid waste crisis of unprecedented scale. Current assessments indicate that national waste generation has reached approximately 170,338 tonnes per day (TPD), with only 91,512 TPD undergoing scientific treatment. The remaining deficit contributes to overflowing landfills, environmental degradation, and significant public health hazards. In response, the Government of India, through the Swachh Bharat Mission-Urban 2.0 (SBM-U 2.0) and the Solid Waste Management Rules 2026, has mandated a strict four-way source segregation protocol encompassing wet, dry, sanitary, and domestic hazardous waste. However, the efficacy of this policy is severely undermined by a pervasive cognitive gap at the household level, where citizens struggle to accurately categorize increasingly complex packaging materials.

This comprehensive research report outlines the strategic, technical, and design specifications for a product tentatively named EcoSort AI, conceived for the "Technology for a Cleaner Bharat" AI-first product builder hackathon. The proposed solution is a highly accessible, mobile-first web application that leverages state-of-the-art multimodal large language models (MLLMs) to provide real-time, zero-shot visual waste classification. By allowing users to photograph a waste item and immediately receive localized segregation instructions, EcoSort AI serves as a personalized behavioral intervention tool. Designed under a severe two-hour implementation constraint, the product architecture prioritizes rapid deployment using Next.js, Vercel Serverless Functions, and the @google/genai SDK, intentionally bypassing complex database requirements in favor of a flawless, edge-optimized core user journey. The ensuing sections critically evaluate the solid waste ecosystem in India, justify the selected product concept against competing alternatives, and provide an exhaustive, implementation-ready blueprint for an AI coding agent.

2. The Exact Problem Statement

The central impediment to effective municipal solid waste management in India is the systemic failure of segregation at the source—primarily the household. While collection efficiency across urban local bodies (ULBs) has reached 95.4%, the treatment and scientific disposal of this waste lag significantly, largely because the incoming waste is heavily commingled. When biodegradable organic matter, which constitutes approximately 51.3% of India's solid waste, is mixed with recyclables (17.48%) and hazardous materials, the entire batch becomes contaminated and economically unviable for material recovery facilities (MRFs).

The fundamental research question investigates why households fail to segregate waste correctly despite widespread civic education campaigns. The analysis reveals that the primary barrier is cognitive friction exacerbated by the increasing complexity of modern consumer goods. Packaging formats have evolved beyond simple paper and glass into multi-layered

plastics (MLPs), composite cartons, and hybrid materials that defy intuitive categorization. For instance, a citizen may reasonably assume a metallic-lined snack wrapper belongs in the dry recyclable bin, whereas standard municipal guidelines often classify such non-recyclable MLPs differently based on local processing capabilities.

Furthermore, recycling instructions provided on packaging are frequently confusing, unreliable, or entirely absent. Standardized global recycling symbols do not reflect the localized realities of Indian municipal infrastructure or the economic drivers of the informal waste sector. The "kabadiwala" (informal scrap dealer) network operates strictly on economic margins, readily accepting high-value materials such as copper wire or clear PET bottles, but rejecting low-value or mixed plastics. Consequently, citizens lack immediate, contextual feedback when faced with ambiguous disposal decisions.

Safety represents another critical dimension of this problem. Domestic hazardous waste, including expired pharmaceuticals, broken glass, and lithium-ion batteries, is frequently discarded in general waste streams. This practice poses severe occupational hazards to sanitation workers and risks catastrophic landfill fires. The absence of a real-time, low-friction tool that can bridge the knowledge gap between complex municipal regulations and everyday household disposal decisions presents a prime opportunity for an AI-first software intervention. A small, rapidly deployable multimodal AI application can directly intervene at the moment of disposal, reducing cognitive load by translating visual input into actionable, safe segregation instructions.

3. Research Evidence and Credible Citations

The conceptualization and architectural design of EcoSort AI are deeply grounded in verified data regarding India's waste management ecosystem, regulatory frameworks, and technological constraints.

The baseline statistics for solid waste generation highlight the scale of the crisis. The Central Pollution Control Board (CPCB) documented that India generated 160,038.9 TPD in the 2020-21 period, which has subsequently been updated to an average of 170,338 TPD. While collection networks are extensive, processing capabilities vary drastically across jurisdictions. For example, states like Chhattisgarh report 100% waste treatment, whereas others lag significantly, demonstrating the need for localized waste intelligence. The composition of this waste is notably different from Western demographics, comprising predominantly biodegradable organic matter alongside a growing fraction of complex recyclables.

Waste Category	Proportion of Total MSW in India	Primary Disposal Challenge
Biodegradable Organic	51.3% ± 8.3%	High moisture content causes severe contamination when mixed with dry waste.
Recyclables (Dry)	17.48% ± 5%	Material ambiguity (e.g., MLPs) prevents accurate source segregation.
Ash and Inert Materials	30% – 40%	Often illegally dumped as construction and demolition (C&D) waste.
Paper and Glass	3% – 6%	Cross-contamination ruins recycling value; broken glass

Waste Category	Proportion of Total MSW in India	Primary Disposal Challenge
		poses safety risks.

To combat this, the Swachh Bharat Mission-Urban 2.0 (SBM-U 2.0) guidelines enforce a mandatory four-way segregation model: Wet Waste, Dry Waste, Sanitary Waste, and Domestic Hazardous Waste. This policy shift from basic two-bin systems requires a massive behavioral adjustment from the public. Furthermore, the Extended Producer Responsibility (EPR) framework under the Plastic Waste Management Rules imposes strict collection and recycling targets on producers, importers, and brand owners for four distinct categories of plastic packaging. However, these macro-level policies fail to address the micro-level confusion at the household bin.

The economics of the informal recycling sector further complicate the landscape. The rates offered by local scrap dealers dictate the flow of recyclables. High-value materials such as copper (fetching up to ₹600/kg) and clean aluminum (₹95–₹115/kg) are aggressively diverted from landfills, whereas mixed paper (₹6–₹10/kg) and mixed hard plastics (₹8–₹14/kg) are often ignored. Therefore, a technological solution must not merely state if an item is technically recyclable, but rather instruct the user on how to prepare it (e.g., washing and flattening) to ensure it retains value for the informal sector.

Finally, the industrial shift toward Waste-to-Energy (WtE) and Refuse Derived Fuel (RDF) plants necessitates higher quality, high-calorific dry waste. The CPCB has recently reclassified WtE plants and issued strict guidelines regarding their operation and emissions, which are directly impacted by the moisture content of the incoming feed. High moisture, caused by poor segregation of wet waste, severely degrades incinerator efficiency. Thus, enabling accurate source segregation via consumer technology directly supports national infrastructure objectives.

4. Comparison of at least five product concepts

To determine the optimal product direction for the two-hour hackathon constraint, five distinct AI-first concepts were evaluated across the six provided tracks. The evaluation prioritized the severity of the problem addressed, technical feasibility, and the strength of the functional AI component.

The first concept evaluated was a **Camera-Based Waste Identification and Segregation Guide**. This solution relies on multimodal large language models to instantly classify a user-uploaded image of waste into the mandated municipal categories and provide precise disposal instructions. This directly addresses the cognitive load problem at the source. The AI component is central and highly functional, leveraging advanced vision capabilities to parse complex materials. Given the availability of robust APIs, this is highly feasible for a rapid MVP.

The second concept considered was a **Locality-Aware Disposal and Recycling Discovery Directory**. This application would use geolocation to map nearby authorized recyclers, e-waste drop-off points, and municipal bins. While highly useful for bulky or hazardous waste, the functional AI component is weak, often relegated to basic search optimization. Furthermore, gathering and verifying localized geospatial data for informal and formal recyclers across India is impossible within a two-hour window, rendering the MVP unfeasible.

A **Waste-Recycling Marketplace Connecting Households to Local Recyclers** was the third concept. This peer-to-peer platform would attempt to formalize the kabadiwala network, allowing users to list household waste for pickup. However, the economic reality invalidates this for daily waste; the scrap value of a single plastic bottle or cardboard box does not justify the logistical

cost of a targeted pickup. Building the necessary dual-sided authentication, database schema, and matching algorithms far exceeds the emergency MVP timeframe.
The fourth concept was an **AI-Assisted Community Litter Reporting System**. Users would photograph illegal dumping sites, and the AI would estimate the volume and categorize the waste, routing a priority ticket to the municipality. While this utilizes vision AI effectively, it addresses the symptom (litter) rather than the root cause (improper household disposal). Additionally, it requires complex integrations with existing municipal grievance redressal systems, making it difficult to demonstrate a complete, closed-loop user journey in a standalone hackathon pitch.
Finally, a **Gamified Waste Education and Behaviour-Change System** was evaluated. This would involve AI-generated quizzes and interactive sustainability modules. While technically trivial to implement, it lacks a compelling, real-world utility during the actual moment of waste disposal. Educational apps suffer from low retention and fail to assist a user holding a confusing piece of packaging in real-time.

Evaluation Criteria	Camera-Based Guide	Recycling Directory	P2P Marketplace	Litter Reporting	Gamified Education
Originality & Differentiation	High	Low	Medium	Low	Low
Problem Severity Addressed	High (Source Segregation)	Medium	Medium	Medium	Low
Strength of AI Component	High (Vision/MLLM)	Low (Search)	Low (Matching)	High (Vision)	Low (Generative Text)
Feasibility (2-Hour MVP)	**Very High**	Low (Data constraint)	Low (Complexity)	Medium	High
Impact Potential	High	Medium	Low	Medium	Low

The evaluation matrix definitively points to the Camera-Based Waste Identification Guide as the most viable, impactful, and technically impressive concept for the hackathon parameters.

5. Selected Solution and Differentiation

Based on the comparative analysis, the selected product concept is **EcoSort AI**, an AI-powered waste intelligence assistant designed to operate seamlessly via a mobile web browser. EcoSort AI directly tackles the core challenge of cognitive friction during source segregation by answering four critical questions for the user in real-time: the item's identity, its municipal waste category, safe preparation instructions, and recommended disposal actions.
It is crucial to acknowledge that EcoSort AI is not entirely unprecedented. Industrial solutions exist, such as TrashCon's TrashBot, which utilizes AI to operate physical, mechanical segregators capable of processing up to 200 tonnes of mixed waste per day. Similarly, Saahas Zero Waste collaborates with Google's CircularNet AI to improve plastic waste sorting accuracy at material recovery facilities. Furthermore, platforms like Recykal provide digital marketplaces that formalize the recycling ecosystem and assist brands with EPR compliance.
However, EcoSort AI's differentiation strategy is rooted in its hyper-accessible, consumer-facing behavioral intervention model. Existing solutions overwhelmingly target the B2B sector, focusing on mechanical sorting *after* the waste has already been commingled, or they focus on

supply-chain logistics. EcoSort AI operates at the very beginning of the waste lifecycle: the consumer's hand. By prioritizing a friction-free, no-download, no-login web experience, the product democratizes advanced waste intelligence. It empowers the ordinary citizen to comply with the SBM-U 2.0 four-way segregation mandate without requiring them to memorize complex polymer classifications or local municipal by-laws. The product is not merely a classifier; it is a personalized compliance and safety assistant.

6. Target Users and Core User Journeys

The design and functionality of EcoSort AI are tailored to a diverse demographic navigating the complexities of the Indian waste management system. The primary target users include environment-conscious urban households striving to comply with municipal segregation mandates, students and educational institutions participating in civic sustainability programs, and facility management staff in residential societies who require a rapid reference tool for ambiguous items.

The product relies on a seamless, frictionless core user journey that guides the individual from uncertainty to correct action in a matter of seconds. The journey must account for varying levels of digital literacy and device capability.

The Core User Journey:

1. **The Trigger:** A user prepares to dispose of an item—for example, a partially full aerosol can or a heavily soiled food delivery container—and realizes they are unsure whether it belongs in the dry recyclable bin, the wet organic bin, or requires special handling.
2. **Access and Capture:** The user opens the EcoSort AI web application on their smartphone browser. Bypassing any login screens, they immediately tap the primary "Identify My Waste" button. Utilizing HTML5 device API integrations (capture="environment"), the application invokes the native rear-facing camera, allowing the user to snap a clear photograph of the item. On desktop devices, the interface seamlessly adapts to offer a drag-and-drop file upload zone.
3. **Processing and Analysis:** Upon image capture, the client-side application instantly resizes and compresses the image to minimize bandwidth consumption and adhere to serverless architectural limits. The image is transmitted to the backend, where a multimodal AI model analyzes the visual data against a strict prompt framework defining Indian municipal waste categories. During this brief latency, the user is presented with a calming loading state featuring dynamic sustainability facts.
4. **The Resolution:** The application renders a polished result card. The AI provides a probable item name, definitively categorizes it into one of the four mandated streams (e.g., "Domestic Hazardous Waste"), and offers actionable, step-by-step preparation guidance (e.g., "Do not puncture. Keep away from heat sources. Dispose of in the dedicated red hazardous waste bin").
5. **Continued Engagement:** If the user has multiple items, they can easily scan again, with recent queries stored locally in a session history for quick reference.

7. Final MVP Scope and Features to Postpone

Given the extreme constraint of a two-hour implementation window from start to submission, rigorous scope control is the difference between a functional, winning prototype and a broken, incomplete application. The scope is strictly bifurcated to ensure the core AI workflow operates

flawlessly.
A. MUST BUILD NOW (The Emergency MVP): This category contains the non-negotiable features required to demonstrate a complete, working product to the hackathon judges.

- **Responsive Landing Page:** A clean, visually compelling entry point that establishes trust and clearly articulates the product's value proposition without requiring scrolling.
- **Robust Image Upload Workflow:** Implementation of native camera capture on mobile and file selection on desktop, critically paired with client-side image compression to guarantee API stability.
- **Live AI Integration:** A server-side API route that securely communicates with a real multimodal model (Gemini) to perform genuine zero-shot image classification. Simulated or hardcoded responses are strictly prohibited.
- **Actionable Result Screen:** A UI component that gracefully parses the structured JSON response from the AI, displaying the classification, clear reasoning, and explicit disposal instructions.
- **Resilient Error Handling:** Comprehensive validation for file types and sizes, clear loading indicators, and graceful fallback mechanisms if the AI model times out or returns an unparseable response.
- **Static Educational Guide:** A minimal, hardcoded reference section explaining the SBM-U 2.0 four-way segregation categories.

B. BUILD ONLY IF TIME REMAINS: These features enhance the user experience but must not jeopardize the completion of the core workflow.

- **Session-Based Scan History:** Utilizing browser localStorage to display recently scanned items, avoiding the need for backend database architecture.
- **Language Toggle:** A simple UI state switch to display instructions in Hindi, expanding the accessibility of the tool for Indian residents.
- **Shareable Results:** A feature allowing users to download their classification result as an image or share it via native web share APIs to promote civic education.

C. FUTURE ROADMAP ONLY (Strictly Excluded from MVP): Attempting to build these features will result in a failed hackathon submission due to time constraints and architectural bloat.

- **User Accounts and Authentication:** Introduces unnecessary friction for a tool meant to be used instantly at the waste bin.
- **Database Integration:** Managing PostgreSQL or MongoDB schemas for an MVP introduces severe deployment risks and latency.
- **Live Recycler Directories and Geolocation:** Collecting and verifying geographic data for local kabadiwalas and MRFs is impossible within two hours.
- **Reward Economies and Tokenization:** Complex payment gateways or blockchain integrations distract from the core visual AI functionality.

8. AI Model Selection and Technical Rationale

The selection of the underlying AI model is the most critical technical decision for EcoSort AI. The requirement is for a system capable of zero-shot visual question answering and robust image classification across an infinite variety of household waste items, many of which may be dirty, damaged, or obscured.
While open-source datasets like TACO (Trash Annotations in Context) and TrashNet exist, training a custom Convolutional Neural Network (CNN) or fine-tuning a local vision transformer

from scratch is technically unfeasible within a two-hour hackathon. Furthermore, bespoke models trained on limited datasets often suffer a dramatic drop in classification accuracy when exposed to out-of-distribution, real-world scenarios.
Therefore, the architecture demands a highly capable, pre-trained multimodal large language model (MLLM) accessible via a low-latency API. The selected primary model is **Google Gemini 2.5 Flash**, accessed via the newly updated @google/genai Node.js SDK.
Technical Rationale for Gemini 2.5 Flash:

1. **Multimodal Proficiency:** The model natively ingests base64-encoded images alongside complex text prompts, allowing it to perform visual analysis while adhering to strict logical constraints regarding waste categories.
2. **Structured Output Enforcement:** MLLMs are susceptible to the "Garbage In, Garbage Out" phenomenon, where unstructured processing leads to hallucinations or irrelevant verbosity. Gemini 2.5 Flash can be explicitly instructed to return responses in a rigid JSON schema, which is essential for rendering predictable frontend UI components.
3. **Latency and Ecosystem Integration:** The "Flash" variant is optimized for rapid response times, crucial for a consumer application where users will not wait longer than a few seconds. The @google/genai SDK is fully compatible with edge and serverless environments like Vercel.

Model Output Structure (The API Contract): To ensure the frontend receives usable data, the model will be constrained to output a specific JSON payload. The structure must capture the nuance of Indian municipal rules while avoiding false confidence.

- item_name: The probable identification of the object.
- category: Strictly limited to one of the four SBM-U 2.0 streams (Wet, Dry, Sanitary, Domestic Hazardous).
- material: The underlying composition (e.g., PET plastic, organic matter, complex electronics).
- confidence_or_uncertainty: An honest indicator of the model's certainty. If the image is obscured, the model must output a low confidence rating rather than guessing.
- explanation: A concise justification for the categorization.
- preparation_steps: An array of actionable steps (e.g., ["Rinse container", "Flatten to save space"]).
- disposal_instructions: Clear guidance on which bin to utilize.
- safety_warnings: Critical for hazardous items. If the model detects a battery or biomedical waste, this field must trigger prominent UI alerts to prevent the user from disposing of it in general waste.
- local_rules_caveat: A boolean indicating if the user should double-check specific municipal guidelines (e.g., for bulky construction waste).

Fallback Strategy: Relying on external APIs introduces network fragility. If the Gemini API rate-limits the request, times out, or returns a payload that fails JSON parsing, the application architecture must catch the exception. The user will be presented with a graceful error state and automatically redirected to the static Waste Knowledge Guide, ensuring the application remains useful even without active AI connectivity.

9. Data Sources and Dataset Opportunities

While EcoSort AI utilizes a foundational MLLM for the emergency MVP, a credible product roadmap must demonstrate an understanding of domain-specific datasets required for future

model evaluation, fine-tuning, and edge-device deployment.
The premier dataset for this domain is **TACO (Trash Annotations in Context)**. Released under a CC BY 4.0 license, TACO is an open image dataset containing photographs of litter in diverse, real-world environments, complete with manual segmentation masks across a hierarchical taxonomy of 60 categories. Unlike pristine laboratory images, TACO captures waste as it is actually found, making it highly valuable for training robust detection algorithms. A key insight from TACO is the size variability of waste; for instance, cigarettes, a major pollution source, often occupy less than 64x64 pixels in high-resolution images, presenting significant challenges for standard object detection pipelines.
Another foundational dataset is **TrashNet**, which consists of 2,527 images categorized into six broad material classes: glass, paper, cardboard, plastic, metal, and trash. While smaller and less contextually diverse than TACO, TrashNet remains a standard benchmark for evaluating basic convolutional neural networks (CNNs) in waste segregation tasks.
For the hackathon presentation, acknowledging these datasets validates the team's technical depth. The future roadmap (detailed in Section 18) outlines how TACO can be utilized to train lightweight, distilled models (e.g., YOLOv8 or MobileNet variants) for on-device, offline inference, drastically reducing the operational costs associated with continuous LLM API calls.

10. Technical Architecture Diagram

The architecture for EcoSort AI prioritizes speed of implementation, zero-maintenance scaling, and seamless integration between the frontend and the AI processing layer. The system is designed around a serverless computing paradigm.

```
flowchart TD
    %% Frontend Layer
    subgraph Client [User Device / Browser]
        A[Web UI: Next.js / React]
        B[Device Camera / File System]
        C[HTML5 Canvas: Client-Side Compression]
    end

    %% Backend Layer
    subgraph Serverless [Vercel Edge / Serverless Network]
        D[API Route: /api/analyze]
        E[Validation & JSON Parser]
    end

    %% External Services
    subgraph AI_Provider [Google Cloud]
        F[Gemini 2.5 Flash Model]
    end

    %% Flow
    A -->|1. Trigger Upload| B
    B -->|2. Raw Image Data| C
    C -->|3. Compressed Base64 Payload| D
    D -->|4. Authenticated RPC Call via @google/genai| F
```

```
F -->|5. Raw Text JSON Response| D
D -->|6. Parse & Validate Structure| E
E -->|7. Formatted JSON Payload| A
A -->|8. Render Result Component| A
```

Critical Architectural Constraint: Payload Limits A paramount consideration in this architecture is the handling of image sizes. Modern smartphone cameras routinely produce JPEG images exceeding 5 to 10 megabytes. Vercel Serverless Functions impose a strict 4.5 MB maximum payload limit for request and response bodies. If a user attempts to upload a raw, high-resolution image directly to the /api/analyze endpoint, the Vercel infrastructure will automatically reject the request, returning a 413 FUNCTION_PAYLOAD_TOO_LARGE HTTP error before the code even executes.
To circumvent this limitation without requiring a dedicated storage bucket (like AWS S3), the architecture implements a mandatory client-side compression step. When the user selects an image, the React frontend loads the file into an off-screen HTML5 <canvas> element, resizes the maximum dimensions while preserving aspect ratio, and exports the data as a compressed, base64-encoded JPEG string. This ensures the payload consistently remains under 1 MB, guaranteeing reliable transmission to the serverless endpoint while maintaining sufficient visual fidelity for the Gemini model to perform accurate classification.

11. Technology Stack and Folder Structure

The technology stack is selected to eliminate configuration overhead and maximize developer velocity during the two-hour sprint.
The Selected Stack:

- **Framework:** Next.js (utilizing the App Router). This allows the frontend React components and the backend serverless API functions to co-exist within a single unified repository, streamlining deployment.
- **Language:** TypeScript. The strict typing prevents runtime errors when passing complex JSON structures between the API and the UI components, which is critical when dealing with dynamic LLM outputs.
- **Styling:** Tailwind CSS. Utility-first CSS enables rapid, responsive UI development directly within the JSX, eliminating the need for context-switching to separate stylesheet files.
- **AI Integration:** @google/genai SDK. The official, newly released Google SDK designed specifically for the latest Gemini models, providing robust methods for handling multimodal inputs.
- **Deployment:** Vercel. Provides zero-configuration continuous deployment straight from the GitHub repository, automatically provisioning the serverless infrastructure.
- **Icons:** Lucide React. A lightweight, consistent SVG icon library.

Folder Structure (Next.js App Router Paradigm): ecosort-ai/ ├── src/ │ ├── app/ │ │ ├── api/ │ │ │ └── analyze/ │ │ │ └── route.ts # Backend Serverless Function (Gemini integration) │ │ ├── globals.css # Tailwind directives and global styles │ │ ├── layout.tsx # Root layout, meta tags, and font definitions │ │ └── page.tsx # Main SPA Entry (Manages UI state machine) │ ├── components/ │ │ ├── Header.tsx # Branding and navigation │ │ ├── Scanner.tsx # Camera input and Canvas compression logic │ │ ├── ResultCard.tsx # Renders the parsed AI JSON response │ │ └── ErrorBoundary.tsx # Graceful UI degradation on failure │ └── lib/ │ ├── utils.ts # Helper functions (e.g., base64 conversion) │ └── prompts.ts #

Centralized system prompts for the AI ├── public/ │ └── assets/ # Static assets (logos, placeholders) ├── tailwind.config.ts # Custom design tokens (colors, typography) ├── package.json └── .env.local # Environment variables (GEMINI_API_KEY)
Environment Variable Management: The application requires a single secret: the GEMINI_API_KEY. This key must never be exposed to the client-side bundle. By placing it in .env.local and strictly accessing it only within the src/app/api/analyze/route.ts file (which executes exclusively on the server), the architecture ensures complete credential security.

12. UI/UX Design System

The visual identity of EcoSort AI must transcend the generic, unpolished aesthetics often associated with hackathon prototypes. It must present as a credible, premium environmental technology product. The design language emphasizes calmness, trustworthiness, and exceptional readability, utilizing nature-inspired tones anchored by high-contrast accessibility standards.
Color Palette (Design Tokens): The interface avoids stark, clinical whites and aggressive primary colors, opting instead for a sophisticated, earthy palette.

Token Name	Hex Code	Usage
Primary Background	#F7F6EF	Warm Ivory. Serves as the base application background, reducing eye strain and evoking a natural feel.
Brand & Primary Text	#12372A	Forest Green. Used for main headings and dense typography, ensuring WCAG AAA contrast ratios against the ivory background.
Primary Action	#16845B	Emerald. The core call-to-action color for buttons (e.g., "Scan Waste"). Highly inviting and distinct.
Success / Accent	#C8F169	Lime Accent. Used sparingly to highlight key metrics, icons, or successful classification badges.
Card Background	#FFFFFF	Pure White. Provides a clean container for results, lifting them off the ivory background via subtle shadows.
Secondary Text	#69756D	Muted Grey. Used for explanatory text, footers, and minor UI elements where lower visual hierarchy is required.
Critical Warning	#C74646	Error Red. Strictly reserved for safety warnings (e.g., detecting hazardous e-waste or batteries) and system errors.

Typography and Layout Principles:

- **Typography:** The application will utilize a modern, highly legible sans-serif font stack (e.g., Inter or system-ui). Font weights are strictly managed to establish a clear visual hierarchy: bold headings, medium primary buttons, and regular body text.
- **Component Styling:** UI components, specifically the result cards and input areas, will feature a uniform border radius (rounded-2xl or 16px in Tailwind) to create a soft, friendly interface. Drop shadows are restrained (shadow-sm transitioning to shadow-md on hover) to avoid excessive skeuomorphism or distracting glassmorphism effects.
- **State Deliberation:** The design system mandates distinct visual treatments for the application's four core states: Empty (inviting the user to scan), Loading (calming animations with educational text), Success (clear, structured data presentation), and Error (apologetic, offering immediate retry pathways).

13. Mobile and Desktop Responsive Specifications

EcoSort AI is engineered with a strict mobile-first philosophy. The primary use case anticipates a user standing near a waste receptacle, operating their smartphone with one hand. Consequently, the interface must prioritize immediate accessibility to the camera and highly legible output that requires no pinch-to-zoom.

Mobile Experience Specifications:

- **Layout:** A single-column structure utilizing vertical scrolling. Padding is generous (typically 16px to 24px) to prevent content from crowding the screen edges.
- **Input Handling:** The core action relies on the HTML5 input attribute: <input type="file" accept="image/*" capture="environment" />. On mobile browsers, this specific markup bypasses the generic file system picker and directly launches the device's rear-facing camera, drastically reducing the friction required to analyze an item.
- **Touch Targets:** The primary "Scan Waste" button must span the full width of the container (w-full) and maintain a minimum touch height of 48px to comply with mobile accessibility standards.

Desktop Experience Specifications:

- **Layout:** While functional on desktop, the layout adapts using CSS Grid or Flexbox to utilize the wider viewport. At the md: or lg: breakpoints, the application transitions to a balanced two-column layout.
- **Input Adaptation:** Recognizing that desktop computers and laptops generally lack rear-facing cameras suitable for photographing external objects, the interface automatically de-emphasizes the live camera capture. Instead, it prominently features a drag-and-drop file upload zone, allowing users to analyze saved images downloaded from the internet or transferred from other devices. The results card occupies the adjacent column, allowing for side-by-side comparison of the uploaded image and the AI analysis.

Across all platforms, the application must support robust keyboard navigation, visible focus indicators for tabbed browsing, and semantically correct HTML elements to ensure accessibility for users relying on screen readers.

14. Screen-by-Screen Interface Requirements

As a Single Page Application (SPA), EcoSort AI dynamically transitions between distinct interface states without requiring full page reloads, ensuring a fluid user experience. The

application is logically structured into four primary views.

A. The Landing / Empty State

- **Header:** Features the EcoSort AI branding in Forest Green.
- **Hero Section:** A concise, factual headline (e.g., "Segregate with Confidence") followed by a brief sub-headline explaining the app's purpose.
- **Primary Action Area:** A massive, pulsing Emerald button dominating the center of the screen, bearing a camera icon and the text "Identify My Waste."
- **Educational Context:** Below the fold, a simple three-step infographic illustrates the process: 1. Snap a Photo. 2. AI Analyzes Material. 3. Dispose Properly. A minimal footer houses the required privacy notice.

B. The Scanning / Loading State

- **Visual Feedback:** Once an image is captured or selected, a blurred preview is displayed in the background.
- **Progress Indicator:** A centralized, smooth loading spinner takes focus.
- **Dynamic Messaging:** To mitigate perceived latency during the AI analysis (which may take 2–5 seconds depending on network conditions), the UI cycles through educational sustainability facts (e.g., "Did you know? Mixing wet and dry waste renders recyclables useless.").

C. The AI Results State

- **Identification:** The uploaded image is displayed as a thumbnail alongside a bold heading stating the identified item_name.
- **Categorization Badge:** A highly visible, color-coded badge indicating the specific SBM-U 2.0 category (e.g., a green badge for "Wet Waste," a blue badge for "Dry Recyclable").
- **Actionable Instructions:** A structured list displaying the preparation_steps (e.g., "1. Empty remaining food. 2. Rinse lightly. 3. Crush to save space."). This is followed by explicit disposal_instructions.
- **Safety & Warnings:** If the AI detects hazardous materials (e.g., a battery, a fluorescent bulb), an Error Red warning box appears prominently, overriding general disposal advice and demanding specific e-waste handling.
- **Next Action:** A clear secondary button inviting the user to "Scan Another Item," which resets the state machine.

D. The Waste Knowledge Guide (Static Fallback)

- A clean, accordion-style menu accessible via the header.
- It details the four official categories: Wet/Organic Waste, Dry Recyclable Waste, Sanitary Waste, and Hazardous Domestic Waste, providing examples for each. This serves as a vital educational resource and a necessary fallback if the AI service experiences an outage.

15. AI Prompt and API Contract

The reliability of EcoSort AI hinges entirely on the quality of the system prompt supplied to the multimodal large language model. The prompt must strictly bound the AI's behavior, preventing it from hallucinating non-existent categories, offering unsafe advice, or producing verbose, unparseable text. The backend will interact with the Gemini 2.5 Flash model, which has demonstrated strong capabilities in following complex instructions and generating structured outputs.

System Prompt Engineering: The prompt provided to the @google/genai SDK must be

engineered with explicit directives. It establishes the persona, defines the exact municipal rules it must adhere to, and dictates the precise JSON schema required by the frontend application.
Conceptual Prompt Structure:
"You are EcoSort AI, a highly accurate environmental waste categorization assistant specialized in the Indian municipal solid waste system. Your task is to analyze the provided image of a waste item.
CRITICAL RULES:

1. You must categorize the item into exactly ONE of these four official Swachh Bharat Mission streams: 'Wet Waste' (biodegradable), 'Dry Waste' (recyclable), 'Sanitary Waste', or 'Domestic Hazardous Waste'.
2. If the item is heavily mixed, obscured, or unclear, you must state your uncertainty honestly. Do not guess.
3. NEVER provide unsafe instructions for e-waste, lithium-ion batteries, chemicals, or biomedical waste.
4. You must output your analysis ONLY as valid, raw JSON. Do not wrap the JSON in markdown code blocks (e.g., do not use ```json). Return the JSON object directly.

REQUIRED JSON SCHEMA: { "item_name": "String. The common name of the object.", "category": "String. Must be one of the four official streams.", "material": "String. The primary material composition.", "confidence": "String. 'High', 'Medium', or 'Low'.", "explanation": "String. A 1-sentence reason for this classification.", "preparation_steps": ["String", "String"], // Actionable steps before disposal. "disposal_instructions": "String. Clear instruction on which bin to use.", "safety_warnings": "String. Details of any hazards. If none, output 'None'.", "local_rules_caveat": boolean // True if municipal rules often vary for this item. }"
API Contract and Parsing: The Vercel Serverless Function (e.g., route.ts) receives this prompt alongside the base64 image data. Upon receiving the response from Google's servers, the backend must implement defensive parsing. It should attempt to strip any residual markdown formatting that the LLM might stubbornly inject before passing the string to JSON.parse(). If parsing fails, the API must return a structured 500 Error indicating an "AI Parsing Failure," allowing the frontend to display the appropriate graceful degradation UI rather than crashing.

16. Privacy, Safety, and Failure-Handling Design

A consumer-facing application dealing with potentially sensitive user-generated images requires a robust, privacy-first architecture and rigorous safety protocols.
Data Privacy Policy: EcoSort AI operates on a principle of minimal data collection. For the core MVP workflow, the application does not require user accounts, emails, or phone numbers. Crucially, the application does not request device location access, ensuring absolute anonymity during routine use. Images captured by the user are processed ephemerally; they reside in the client's device memory for compression, are transmitted securely over HTTPS to the Vercel serverless function, and are passed to the external AI provider (Google). The application code explicitly does not log, save, or store these images in any database or file system post-analysis. A concise privacy notice in the application footer will clearly communicate that images are sent to an external AI for real-time analysis but are not retained.
Safety and Hazardous Waste Protocols: The most significant real-world risk associated with solid waste is the improper disposal of Domestic Hazardous Waste. Lithium-ion batteries, when crushed in municipal compactor trucks or exposed to the heat of a landfill, frequently cause severe, toxic fires. Similarly, biomedical waste and chemicals pose severe threats to the health

of informal waste pickers.
The AI prompt is specifically engineered to treat safety as paramount. If the model detects a battery, electronic component, or chemical container, it is instructed to populate the safety_warnings field with urgent directives. The frontend design system enforces this safety protocol by rendering any populated safety_warnings string in a high-visibility Error Red alert box, visually overriding the standard disposal instructions and directing the user to seek a designated e-waste or hazardous drop-off point.
Failure Handling Mechanisms: The architecture must anticipate and gracefully handle three primary failure modes:

1. **Payload Size Errors:** As discussed, Vercel restricts payloads to 4.5 MB. If the client-side compression fails and a massive file is transmitted, the server will return a 413 error. The UI must catch this specific network error and instruct the user to try a different image.
2. **API Timeout or Rate Limiting:** Third-party LLM APIs can experience latency spikes. The serverless function must implement a timeout abort controller (e.g., failing gracefully at 10 seconds). The frontend will catch the 504 Gateway Timeout and present a friendly message apologizing for the heavy traffic, alongside a direct link to the static Waste Knowledge Guide.
3. **Model Hallucination / Low Confidence:** If the user uploads a completely black image or an unrecognizable abstraction, the model is prompted to return a "Low" confidence rating. The frontend logic must detect this flag and alter the UI, suggesting the user retake the photo in better lighting rather than displaying nonsensical categorization data.

17. Testing and Evaluation Checklist

Before the hackathon submission deadline, the application must undergo a rigorous manual testing protocol utilizing diverse, real-world imagery to validate the AI's categorization logic and the robustness of the UI state machine.

Test Scenario (Image Input)	Expected Behavior & Categorization	Critical Verification Point
1. Banana Peel or Apple Core	Categorized as **Wet Waste**.	Must instruct the user to place it in the organic/green bin without plastic bags.
2. Clean Amazon Cardboard Box	Categorized as **Dry Waste**.	The preparation_steps must instruct the user to flatten the box to save space.
3. Empty Lays/Snack Wrapper	Categorized as **Dry Waste** (Multi-Layered Plastic).	Must acknowledge the material is MLP. Instructions should advise wiping it clean.
4. Used AA or Lithium Battery	Categorized as **Domestic Hazardous Waste**.	**CRITICAL:** Must trigger the red safety_warnings UI. Must explicitly forbid disposal in general waste.
5. Broken Drinking Glass	Categorized as **Domestic Hazardous Waste**.	Must instruct wrapping the shards securely in thick newspaper to protect sanitation workers.
6. Paper Coffee Cup (Lined)	Categorized as **Dry Waste** (or	The explanation should

Test Scenario (Image Input)	Expected Behavior & Categorization	Critical Verification Point
	municipal dependent).	acknowledge the complexity of the plastic lining vs. paper exterior.
7. Pitch Black Photo or Blur	Returns **Low Confidence** or Error.	The UI must gracefully degrade, stating "Item not recognizable," and offer a retry button.
8. 12 MB High-Res Photo	Successful Upload and Analysis.	Verifies that the client-side HTML5 canvas compression successfully circumvents the 4.5MB Vercel limit.

18. Impact Model and Realistic Future Roadmap

A compelling hackathon submission must articulate a credible theory of change, demonstrating how a small software product can influence systemic issues without relying on exaggerated claims of immediate environmental salvation.

Theory of Change: The current failure in waste segregation is heavily rooted in point-of-action confusion.

- **Intervention:** The user, unsure of how to dispose of an item, photographs it using EcoSort AI.
- **Mechanism:** The AI rapidly processes the visual data, removing the cognitive load of interpreting ambiguous packaging or navigating dense municipal guidelines. It provides a simple, binary instruction.
- **Action:** Empowered with clear knowledge, the user correctly segregates the item into the appropriate bin.
- **Systemic Outcome:** As household compliance increases, the volume of cross-contamination decreases. This directly yields higher-quality organic matter for composting and cleaner, higher-value materials for the informal recycling sector, ultimately reducing the total tonnage of mixed waste sent to overburdened landfills.

Realistic Future Roadmap (Post-Hackathon): To transition from a hackathon MVP to a deployable civic technology, the product roadmap focuses on reducing operational costs and increasing localized utility.

1. **Q3 2026 (Localization & Discovery):** Integrate geolocation APIs to map municipal Material Recovery Facilities (MRFs) and registered e-waste handlers. If a user scans a high-value or hazardous item, the app will dynamically provide routing to the nearest appropriate facility.
2. **Q4 2026 (Edge AI Migration):** The reliance on external LLM APIs (like Gemini) introduces recurring costs and latency. The team will utilize datasets like TACO and TrashNet to fine-tune lightweight, specialized vision models (e.g., YOLOv8 or MobileNet). These models can be deployed directly to the user's device via TensorFlow.js or ONNX Web Runtime, enabling instant, offline waste classification without server costs.
3. **Q1 2027 (Municipal Integration):** Develop a B2B dashboard for Urban Local Bodies (ULBs). By aggregating anonymized scan data (e.g., detecting a high volume of confusing MLP packaging in a specific ward), municipalities can target their civic education campaigns more effectively, directly improving their metrics for the annual Swachh

Survekshan rankings.

19. Two-Hour Build and Submission Priorities

Given a strict two-hour implementation timeline, attempting simultaneous development on the same files will result in catastrophic merge conflicts and integration failure. The team, comprising Daksh Khandelwal and Khushi Kushwah, must divide responsibilities rigidly along the client/server boundary.

Daksh Khandelwal: Frontend Architecture, UX/UI, and State Management

- Initialize the Next.js project and configure the Tailwind CSS design tokens (colors, typography).
- Develop the core UI components: Header, LandingPage, and ResultCard.
- Implement the critical Scanner component, specifically the HTML5 camera capture (capture="environment") and the JavaScript logic for loading the image into a Canvas for compression and base64 encoding.
- Manage the React state machine (Idle -> Loading -> Success/Error) and ensure all loading spinners and error states render beautifully.

Khushi Kushwah: Backend Infrastructure, AI Integration, and Deployment

- Establish the Vercel project environment and secure the .env.local variables (specifically the GEMINI_API_KEY).
- Develop the Next.js API Route Handler (src/app/api/analyze/route.ts).
- Integrate the @google/genai SDK, writing the logic to ingest the base64 image and construct the rigorous system prompt detailed in Section 15.
- Implement the defensive backend logic: stripping markdown from the LLM response, executing JSON.parse(), and handling API timeouts or provider errors to ensure the endpoint never crashes the client.
- Manage the final integration testing, ensuring the frontend successfully POSTs to the backend and receives the formatted JSON payload.

20. Three-Minute Demo Storyboard

The product demonstration video is the primary criteria for hackathon judging. It must be narrative-driven, visually engaging, and prove that the software actually works.

0:00 - 0:45 (The Hook & The Problem):

- *Visual:* Fast-paced montage of overflowing garbage bins and confused consumers holding complex packaging (e.g., a multi-layered plastic chip packet).
- *Audio:* "Every day, India generates over 170,000 tonnes of waste. Despite government mandates for 4-way segregation, our landfills are overflowing. Why? Because citizens are confused at the bin. Is this foil wrapper dry waste or residual? When people don't know, they mix it. And mixed waste is ruined waste."

0:45 - 2:00 (The Solution & Live Demonstration):

- *Visual:* Transition to a screen recording of a smartphone opening the EcoSort AI web application.
- *Audio:* "Meet EcoSort AI. Your personal, pocket-sized Swachh Bharat assistant. No downloads, no accounts. Just instant clarity."
- *Action:* The demonstrator taps 'Identify My Waste', the camera opens, and they snap a photo of a heavily soiled plastic food container. The screen displays the calming loading

state.
- *Visual:* The Result Card snaps into view.
- *Audio:* "In seconds, our multimodal AI identifies the soiled container. But it doesn't just say 'Plastic'. It gives precise, actionable instructions: 'Empty food residue, rinse lightly, and place in the Dry Recyclable bin.' We just saved that entire bin from contamination."

2:00 - 2:40 (Safety and the Edge Case):
- *Visual:* The demonstrator scans a used AA battery. The screen instantly flashes the Error Red safety warning.
- *Audio:* "EcoSort AI is built for safety. If a user tries to throw away a lithium or alkaline battery, the system instantly flags it as Domestic Hazardous Waste, preventing a potential landfill fire and protecting our vital sanitation workers."

2:40 - 3:00 (Conclusion):
- *Visual:* The team members (Daksh and Khushi) alongside a slide detailing the tech stack (Next.js, Vercel, Gemini 2.5 Flash).
- *Audio:* "Built in under two hours using Next.js and Google Gemini, EcoSort AI proves that AI can drive massive behavioral change. Technology for a Cleaner Bharat. Thank you."

21. System-Design Walkthrough Storyboard

The required system-design video must prove technical competence and explain how architectural constraints were overcome.

0:00 - 0:45 (Architecture Overview):
- *Visual:* Display the Mermaid architecture diagram (from Section 10).
- *Audio:* "Welcome to the EcoSort AI system design walkthrough. We architected this application for maximum accessibility and zero-maintenance scaling. It is a serverless Next.js application deployed on Vercel, bridging a mobile-first React frontend with the Google Gemini 2.5 Flash multimodal AI."

0:45 - 1:45 (The Payload Engineering Challenge):
- *Visual:* Highlight the flow between the mobile device and the Vercel Serverless function. Show a code snippet of the Canvas compression.
- *Audio:* "Our biggest technical hurdle was infrastructure limits. Modern phone cameras generate 10-megabyte images, but Vercel serverless functions strictly reject any payload over 4.5 megabytes. To solve this without adding the latency and cost of a dedicated AWS S3 bucket, we engineered a client-side compression pipeline. The browser loads the raw image into an HTML5 Canvas, downscales it, and outputs a lightweight base64 string, guaranteeing our API requests never fail."

1:45 - 3:00 (Prompt Engineering & Data Parsing):
- *Visual:* Show the backend route.ts file and the strict JSON prompt.
- *Audio:* "On the backend, we interface with Gemini using the new @google/genai SDK. Because large language models are prone to unstructured verbosity, we utilized strict prompt engineering. We command the AI to act as an SBM-U 2.0 expert and mandate that it returns only raw, valid JSON matching our exact interface schema. The serverless function defensively parses this JSON, catches any formatting hallucinations, and delivers clean data to the frontend state machine for rendering. Thank you for watching."

22. Submission-Ready 300–500-Word Write-Up

Project Title: EcoSort AI: Your Personal Swachh Bharat Assistant **One-Line Description:** An accessible, mobile-first web application that uses multimodal AI to instantly identify household waste and provide precise, safe segregation instructions.
The Core Challenge: Despite the Swachh Bharat Mission-Urban 2.0 mandate for 4-way source segregation, household compliance remains the critical bottleneck in India's waste management ecosystem. The primary barrier is not citizen apathy, but severe cognitive friction. Modern packaging materials—such as multi-layered plastics (MLPs) and composite cartons—are confusing. When individuals face an ambiguous item at the bin, they default to throwing it in mixed waste. This cross-contamination ruins the economic viability of dry recyclables, degrades the efficiency of Waste-to-Energy incinerators, and perpetuates the crisis of overflowing landfills.
Our Solution: EcoSort AI acts as a highly personalized, behavioral intervention tool that removes the guesswork from waste disposal. Built as an ultra-accessible, mobile-first web application, it requires no app store downloads and no user accounts. A user simply opens the site, taps "Scan," and photographs the confusing item using their device's native camera. Leveraging the zero-shot visual reasoning capabilities of Google's Gemini 2.5 Flash multimodal AI, EcoSort instantly analyzes the object. Crucially, it translates this visual data into actionable municipal compliance. The app returns a structured result card indicating the exact waste category (Wet, Dry, Sanitary, or Domestic Hazardous), actionable preparation steps (e.g., "Empty contents, rinse, and flatten"), and the correct bin for disposal.
Technical Execution & Safety: To ensure a flawless user experience on serverless architecture, we engineered a client-side HTML5 Canvas compression pipeline, entirely bypassing Vercel's strict 4.5MB payload limits. Furthermore, EcoSort AI is designed with rigorous safety guardrails. Through strict prompt engineering, if the AI detects items like lithium-ion batteries, broken glass, or e-waste, it visually overrides general instructions with high-contrast safety warnings, explicitly preventing hazardous materials from entering general waste streams where they threaten sanitation workers.
EcoSort AI demonstrates that small, meticulously designed AI interventions can bridge the gap between complex government environmental policies and everyday citizen action, driving real behavioral change for a Cleaner Bharat.

23. Tools and Libraries Disclosure List

Category	Tool / Library / SDK	Purpose
Frontend Framework	Next.js (React)	Application framework utilizing the App Router paradigm.
Language	TypeScript	Ensures strict type safety for parsed AI JSON payloads.
Styling	Tailwind CSS	Utility-first CSS for rapid, responsive interface design.
Iconography	Lucide React	Lightweight, consistent SVG icon set.
AI Model	Google Gemini 2.5 Flash	Multimodal LLM utilized for zero-shot image classification.
AI Integration SDK	@google/genai (Node.js)	Official SDK for securely invoking the Gemini API on the backend.
Hosting & CI/CD	Vercel	Serverless deployment platform

Category	Tool / Library / SDK	Purpose
		for the Next.js application.
Version Control	Git & GitHub	Source code management and collaboration.

24. GitHub Repository and README Outline

Proposed Repository URL: github.com/dk-khandelwal06/ecosort-ai
README Outline:

- **Header:** Project Title, logo, and a high-resolution mockup image of the mobile interface displaying a successful waste scan.
- **Badges:** Build status, Next.js version, License (MIT).
- **Overview:** A concise one-paragraph summary of the project and a prominent hyperlink to the live deployed Vercel application.
- **Key Features:**
  - Zero-friction, no-login web interface.
  - Native mobile camera integration.
  - Client-side image compression for rapid, low-bandwidth processing.
  - Real-time multimodal AI analysis using Gemini 2.5 Flash.
  - Strict adherence to SBM-U 2.0 four-way segregation categories.
- **Technology Stack:** Bulleted list of tools (Next.js, Tailwind, Vercel, Google Gen AI).
- **Local Installation & Setup:**
  1. git clone https://github.com/dk-khandelwal06/ecosort-ai.git
  2. cd ecosort-ai
  3. npm install
  4. Create a .env.local file in the root directory.
  5. Add your API key: GEMINI_API_KEY=your_actual_key_here
  6. npm run dev (Access via localhost:3000)
- **Hackathon Context:** A brief acknowledgement that the project was built for the "AI First Product Builder Hackathon 2026 - Technology for a Cleaner Bharat."
- **Contributors:** Daksh Khandelwal & Khushi Kushwah (with links to LinkedIn/GitHub profiles).

25. ANTIGRAVITY IMPLEMENTATION HANDOFF

(Provide the following text directly to the AI coding agent, such as Antigravity or Cursor, to generate the application).
SYSTEM DIRECTIVE FOR AI CODER: You are tasked with building the complete MVP for **EcoSort AI**, an emergency hackathon submission. It is a Next.js (App Router) web application that uses multimodal AI to identify waste and provide segregation instructions. You must use React, TypeScript, and Tailwind CSS. The application must be a mobile-first Single Page Application (SPA).
1. DEPENDENCIES & INITIALIZATION:

- Initialize a Next.js project.
- Install required packages: react, react-dom, tailwindcss, lucide-react, and @google/genai.

2. DESIGN TOKENS (Modify tailwind.config.ts): Extend the theme colors to match the premium brand identity:

- brand-bg: #F7F6EF (Warm Ivory)

- brand-dark: #12372A (Forest Green)
- brand-primary: #16845B (Emerald)
- brand-accent: #C8F169 (Lime Accent)
- brand-error: #C74646 (Error Red)

3. FRONTEND IMPLEMENTATION: src/app/page.tsx

- Create a clean, single-page UI with a brand-dark header.
- Implement a React state machine to manage views: idle | loading | result | error.
- **Camera Input:** Use a hidden HTML5 input triggered by a large, brand-primary <button>: <input type="file" accept="image/*" capture="environment" className="hidden" />
- **Crucial Compression Logic:** When the onChange event fires, you MUST load the selected file into an HTML5 <canvas>, resize it so the maximum dimension (width or height) does not exceed 800px, and output it using canvas.toDataURL('image/jpeg', 0.7). This base64 string is mandatory to bypass Vercel's 4.5MB payload limit.
- **Fetch Call:** POST the compressed base64 string to the /api/analyze endpoint.
- **Result UI:** If the state is result, render a card (bg-white, rounded-2xl, shadow-md). Parse and display the JSON fields: item_name, a colored badge for category, a mapped list for preparation_steps, and disposal_instructions.
- **Safety Trigger:** If the safety_warnings string is anything other than "None", render it prominently in a brand-error colored alert box. Provide a "Scan Another" button to reset the state to idle.

4. BACKEND IMPLEMENTATION: src/app/api/analyze/route.ts

- Initialize the Google Gen AI client: import { GoogleGenAI } from '@google/genai'; const ai = new GoogleGenAI({ apiKey: process.env.GEMINI_API_KEY });
- Extract the base64 image string from the POST body. Strip the prefix.
- **The System Prompt:** Use exactly this string: *"Analyze this waste item. Categorize into ONE of these exactly: 'Wet Waste', 'Dry Waste', 'Sanitary Waste', or 'Domestic Hazardous Waste'. NEVER give unsafe advice for batteries or e-waste. Output ONLY valid, raw JSON without markdown code blocks. Schema required: { "item_name": "string", "category": "string", "material": "string", "confidence": "string", "explanation": "string", "preparation_steps": ["string"], "disposal_instructions": "string", "safety_warnings": "string", "local_rules_caveat": boolean }"*
- Call ai.models.generateContent using the model "gemini-2.5-flash". Pass the prompt and the inline image object (mimeType: "image/jpeg").
- **Defensive Parsing:** Strip any Markdown formatting (e.g., ````json) that the LLM might return in response.text`. Parse it via `JSON.parse()`.
- Return the parsed object via NextResponse.json(). Wrap the entire function in a try/catch block; on error, log to console and return a 500 status with a generic error message so the frontend can degrade gracefully.

5. DEPLOYMENT PROTOCOL: Ensure absolutely no secrets or API keys are exposed on the client side. The Gemini initialization must reside strictly within route.ts. The UI must remain perfectly responsive on narrow 320px screens. DO NOT integrate any databases or authentication mechanisms. Focus entirely on engineering a flawless, error-resistant scan-to-result loop. Execute code generation now.

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